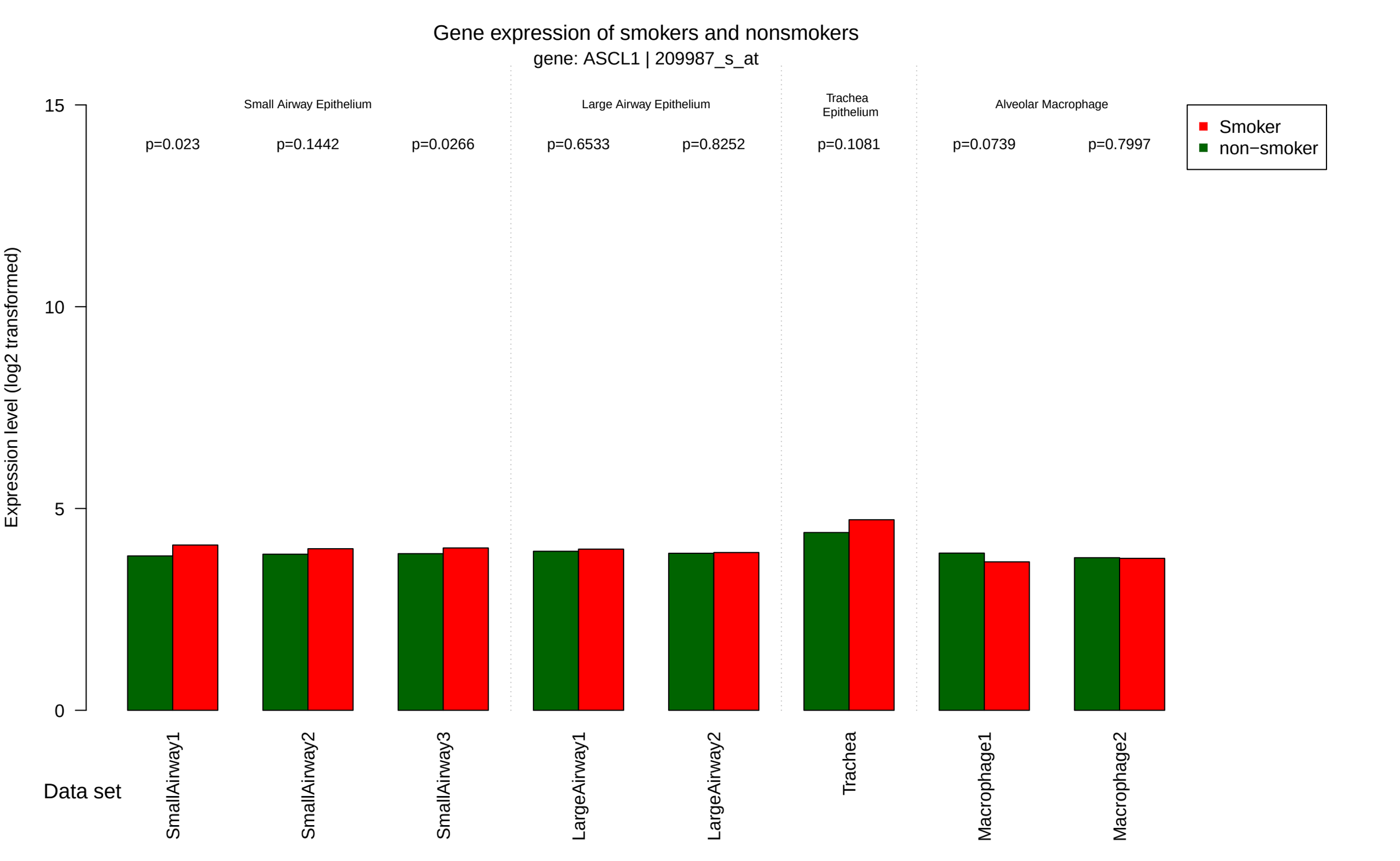
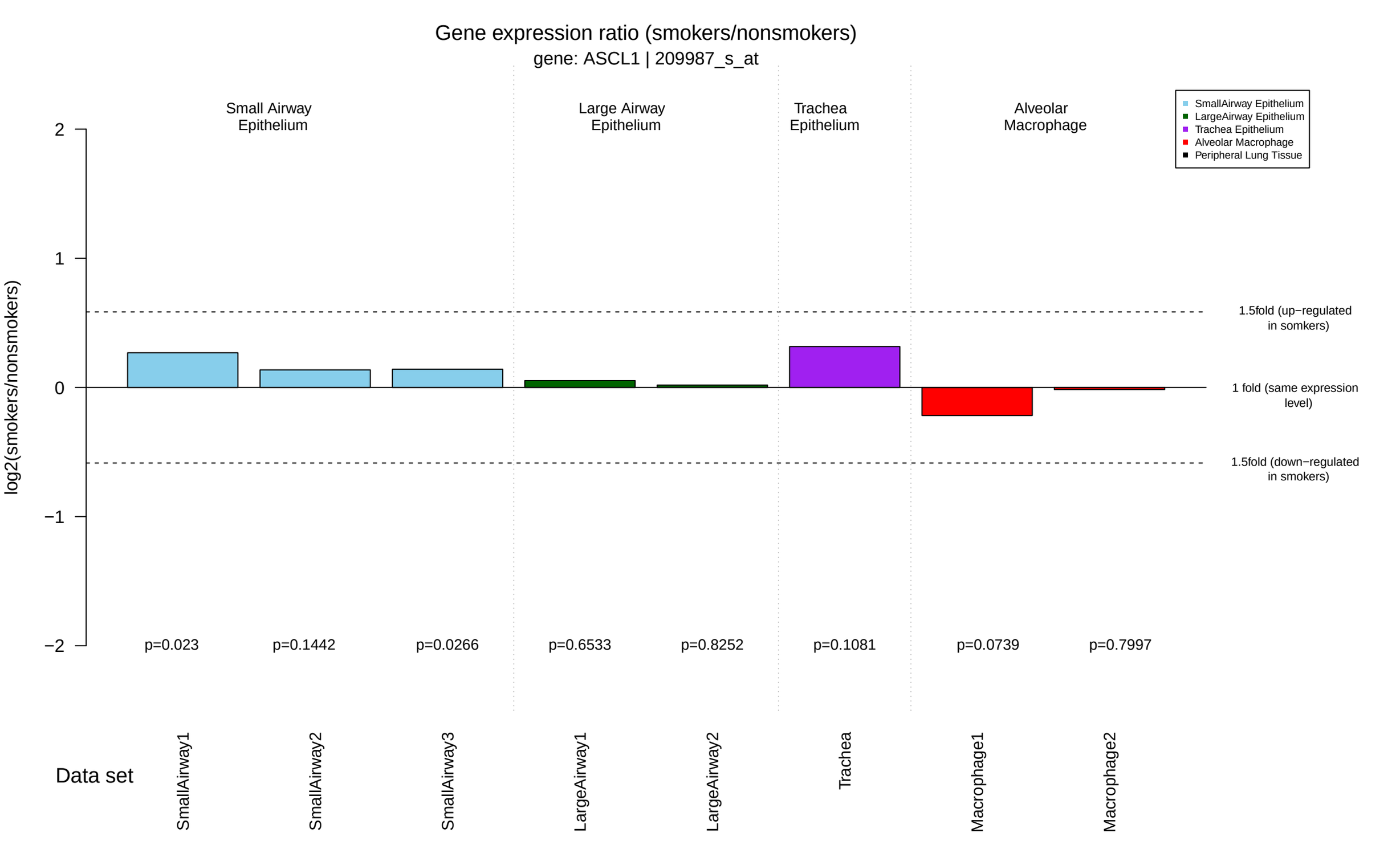
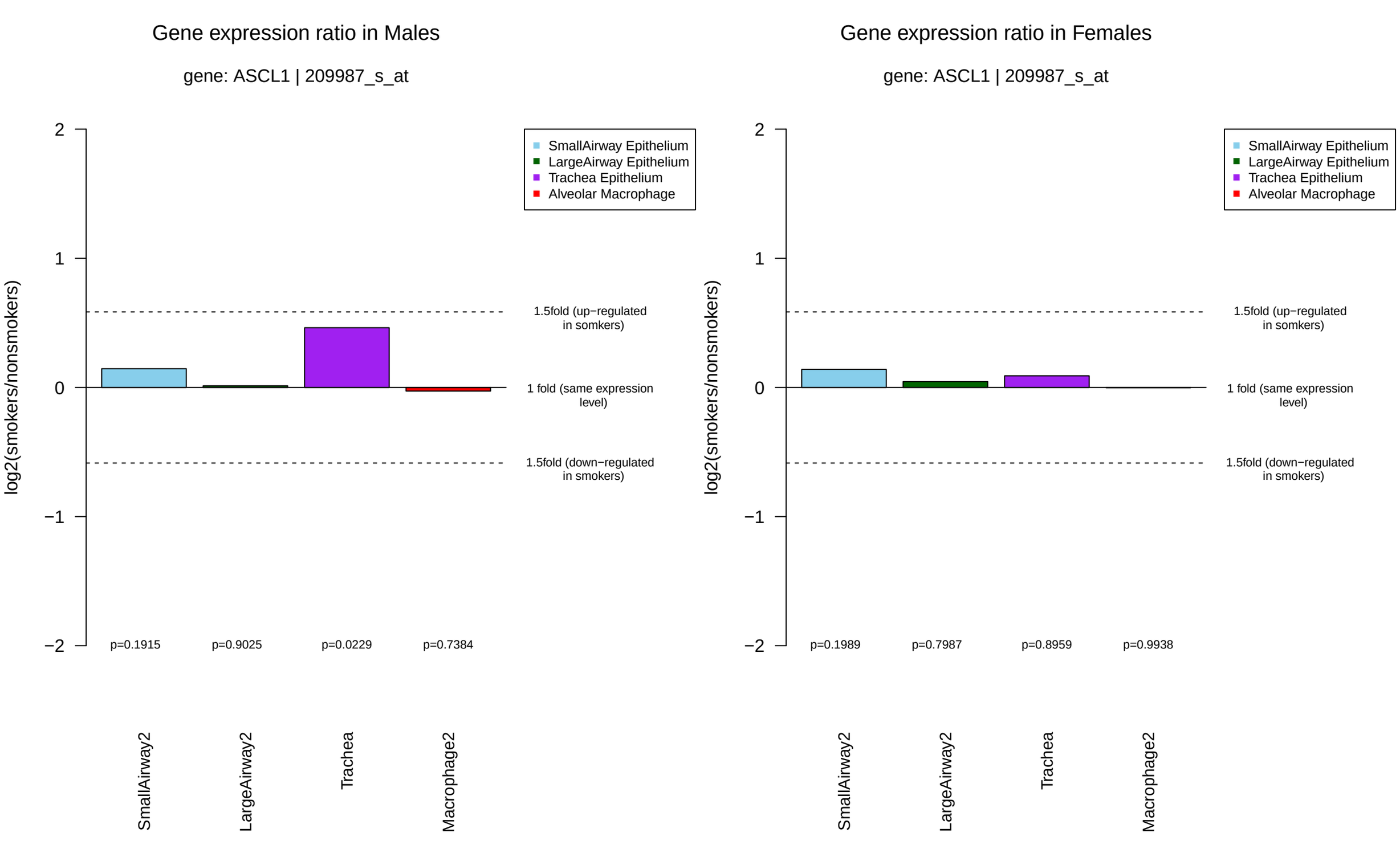
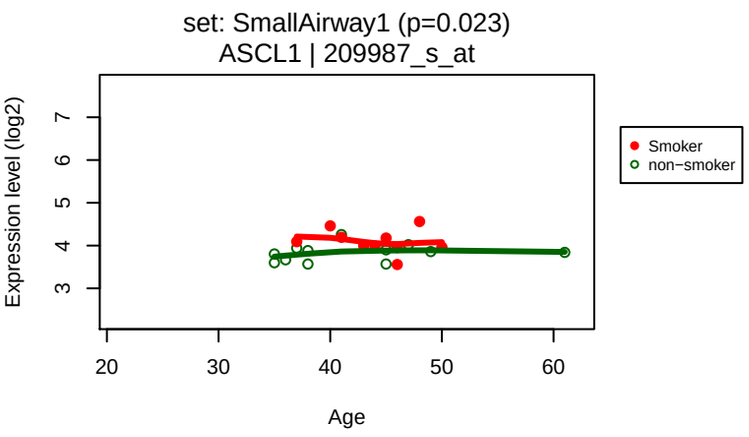
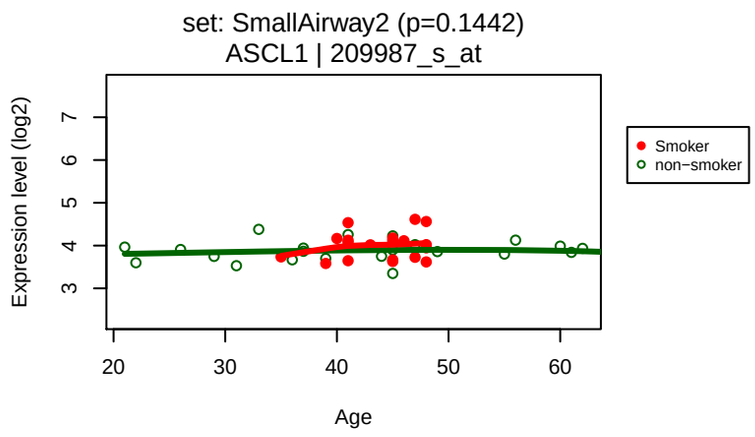
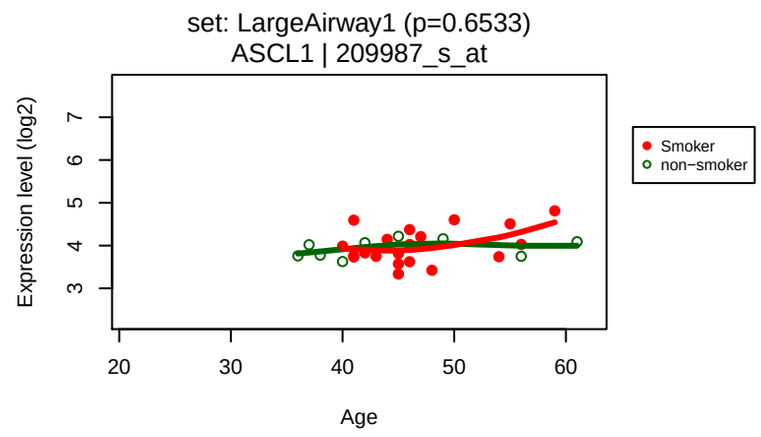

Gene expression of smokers and nonsmokers
gene: ASCL1 | 209987_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.023
p=0.1442
p=0.0266
p=0.6533
p=0.8252
p=0.1081
p=0.0739
p=0.7997
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

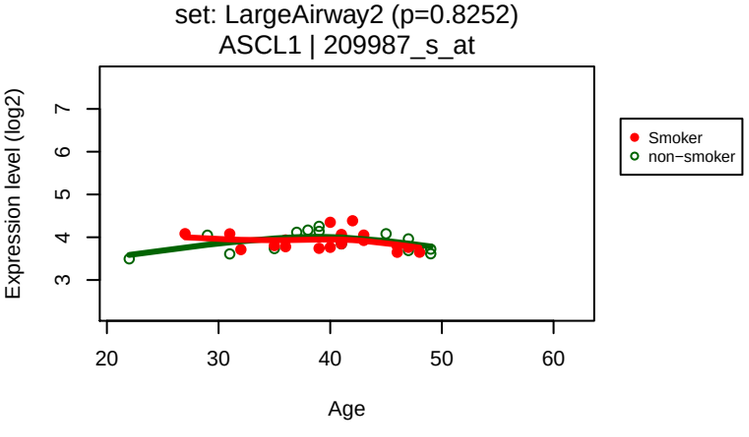

Gene expression ratio (smokers/nonsmokers)
gene: ASCL1 | 209987_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.023
p=0.1442
p=0.0266
p=0.6533
p=0.8252
p=0.1081
p=0.0739
p=0.7997
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

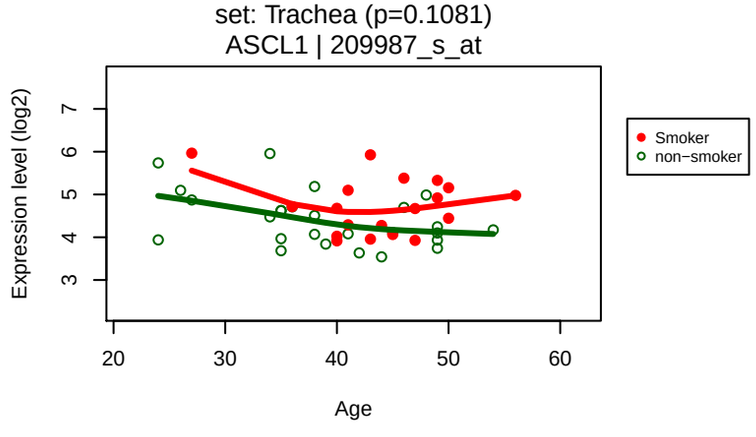

Gene expression ratio in Males
gene: ASCL1 | 209987_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1915
p=0.9025
p=0.0229
p=0.7384
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ASCL1 | 209987_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1989
p=0.7987
p=0.8959
p=0.9938
SmallAirway2
LargeAirway2
Trachea
Macrophage2

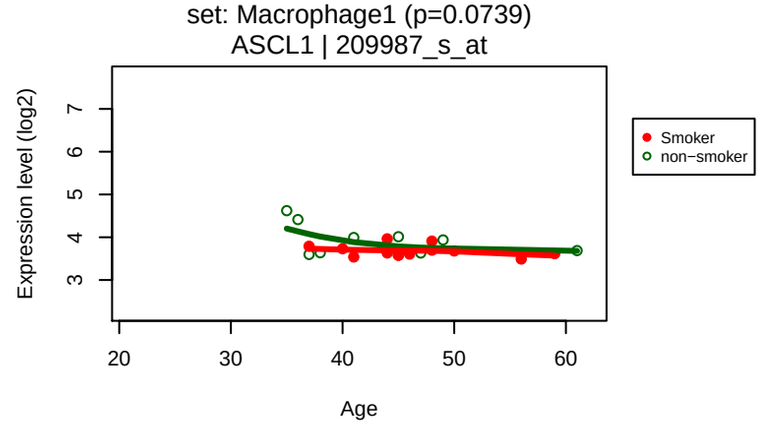

set: SmallAirway1 (p=0.023)
ASCL1 | 209987_s_at
Expression level (log2)
Age
Smoker
non−smoker

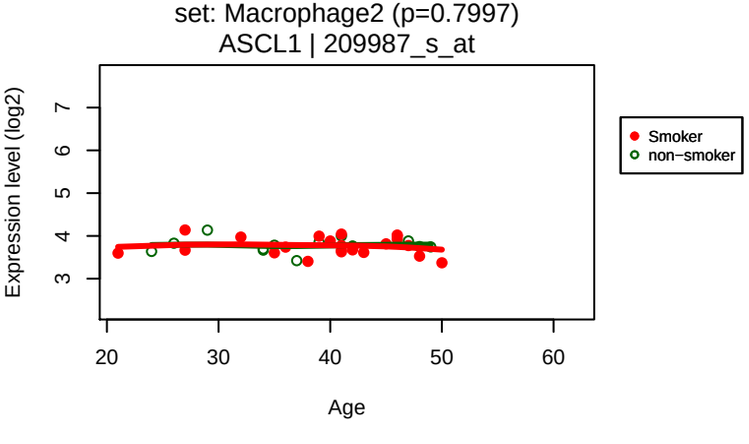

set: SmallAirway2 (p=0.1442)
ASCL1 | 209987_s_at
Expression level (log2)
Age
Smoker
non−smoker

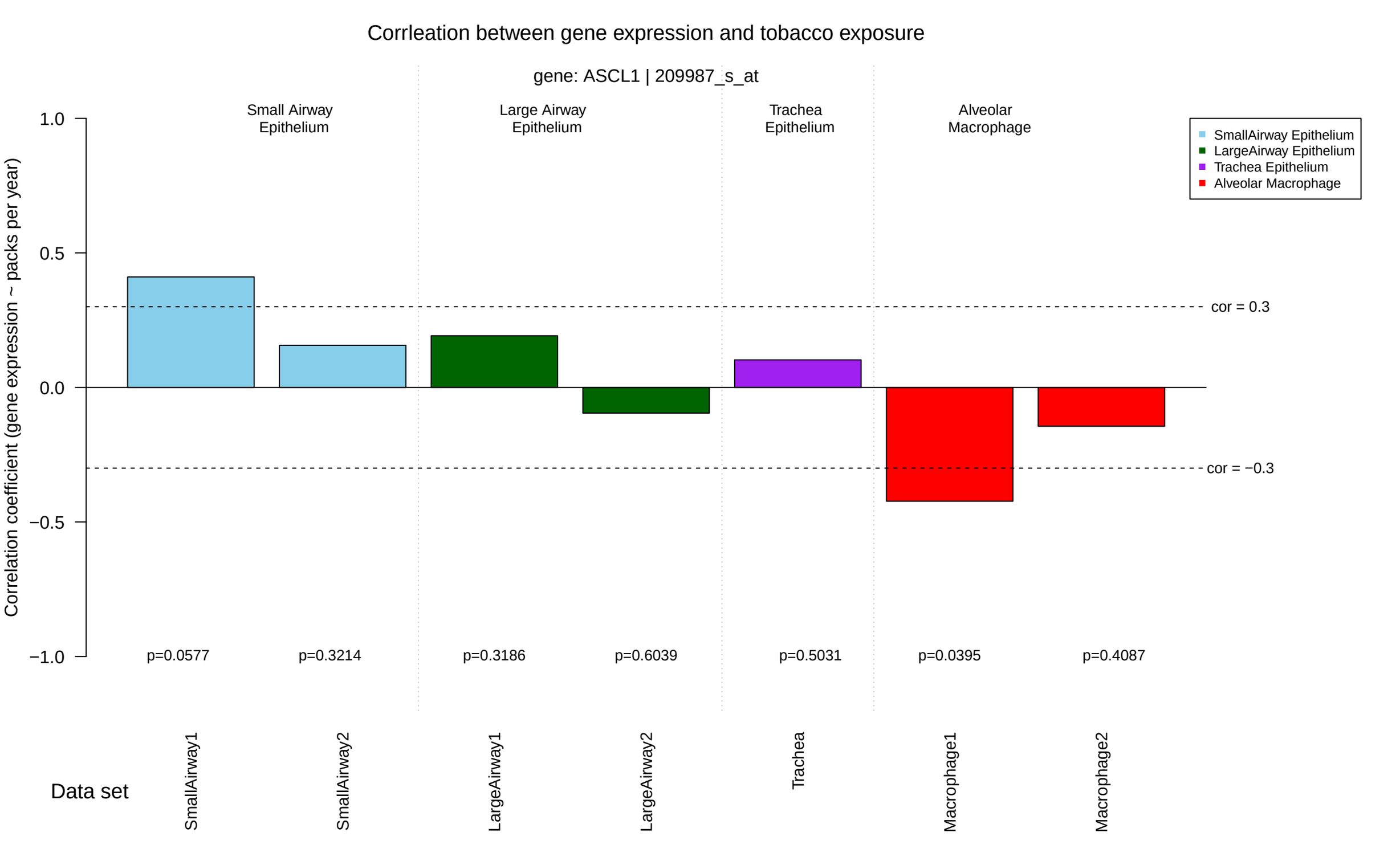

set: LargeAirway1 (p=0.6533)
ASCL1 | 209987_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8252)
ASCL1 | 209987_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1081)
ASCL1 | 209987_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0739)
ASCL1 | 209987_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7997)
ASCL1 | 209987_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ASCL1 | 209987_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0577
p=0.3214
p=0.3186
p=0.6039
p=0.5031
p=0.0395
p=0.4087
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2